

Stochastic Constraint Grouping through Simulated Annealing



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$$P_{\text{acc}} = \begin{cases} 1, & \Delta E \leq 0, \\ \exp\left(-\frac{\Delta E}{T}\right), & \Delta E > 0, \end{cases}$$

$$T \rightarrow \alpha T$$

Dispatch downs

Dispatch Down

1. Curtailment
2. Constraint
3. Surplus



How its done.

Other Countries

1. Nodal Pricing/Locational
Marginal Pricing
2. Zonal pricing



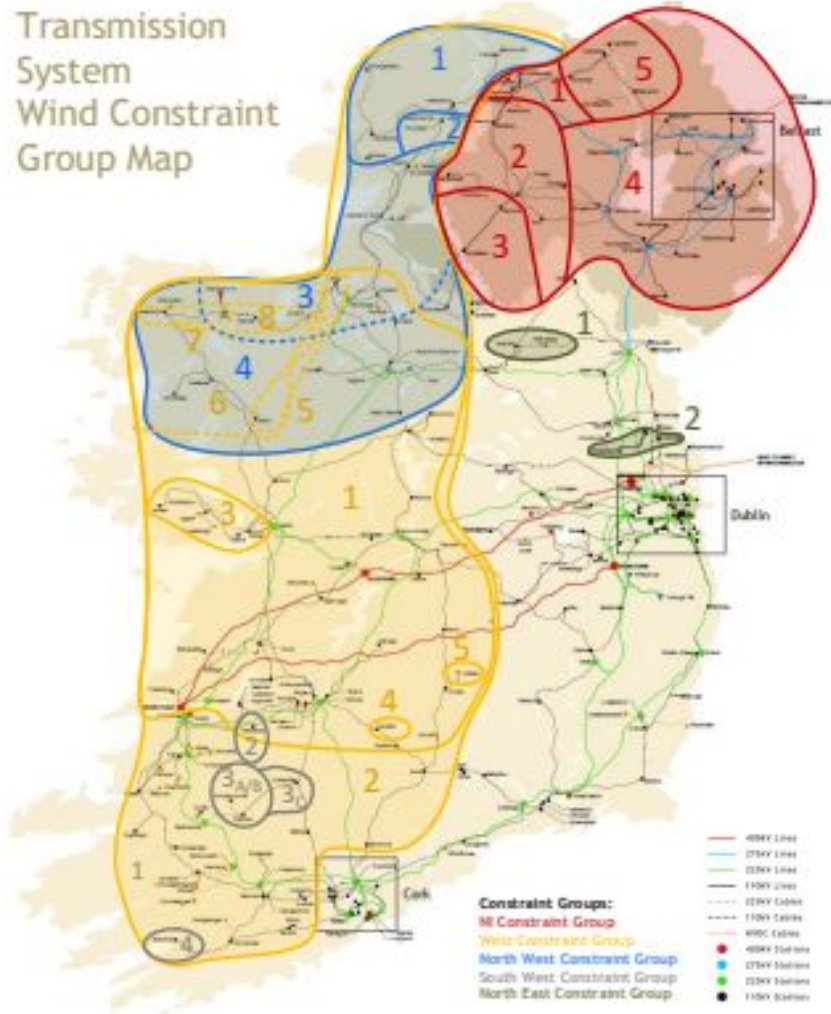
Why not for ireland

- Ireland has a single electricity market where every generator is paid the same wholesale price, meaning the market cannot be used to constrain generation.
- Instead the TSO is in charge of dispatching down generators to satisfy the network's constraints.



Constraint Groups

- Constraint Groups are constructed out of multiple generators which feed into the same transition lines.
- For market fairness all generators in a group are dispatched down equally.



Presentation of the Problem

Static Constraint groups overestimate the amount of the grid which needs to be dispatched down.

Our hypothesis was, that dynamically changing and stochastically spreading the groups will decrease the amount of necessary Dispatch Down.

To do this two things are needed: a pseudo-model of the grid, and an optimisation of the groups.

Modelling Ireland's Grid

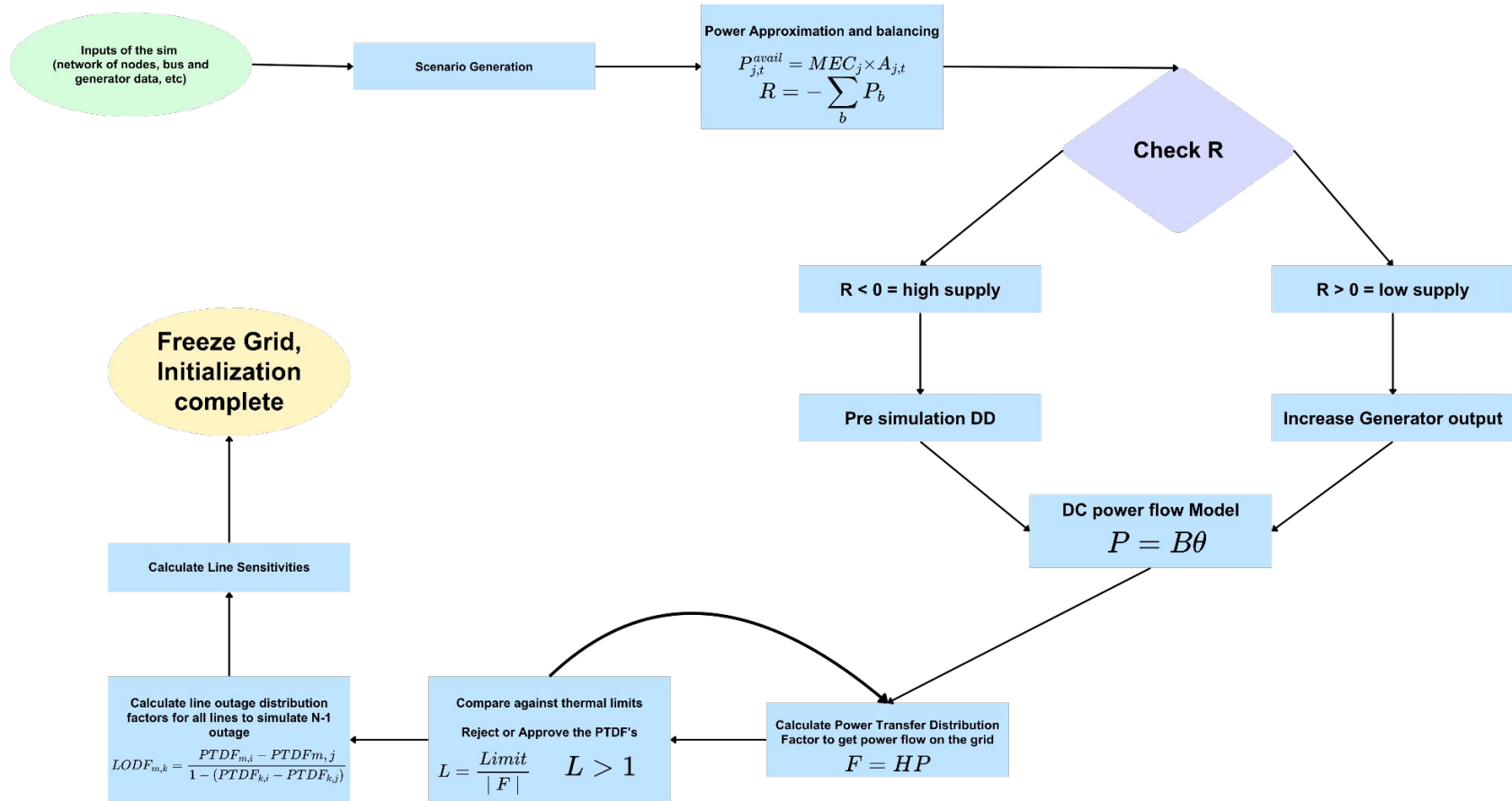
The Grid

The parts of the grid which are important for optimising constraint groups are the renewable energy sector of the grid, and the edges that connect them all together

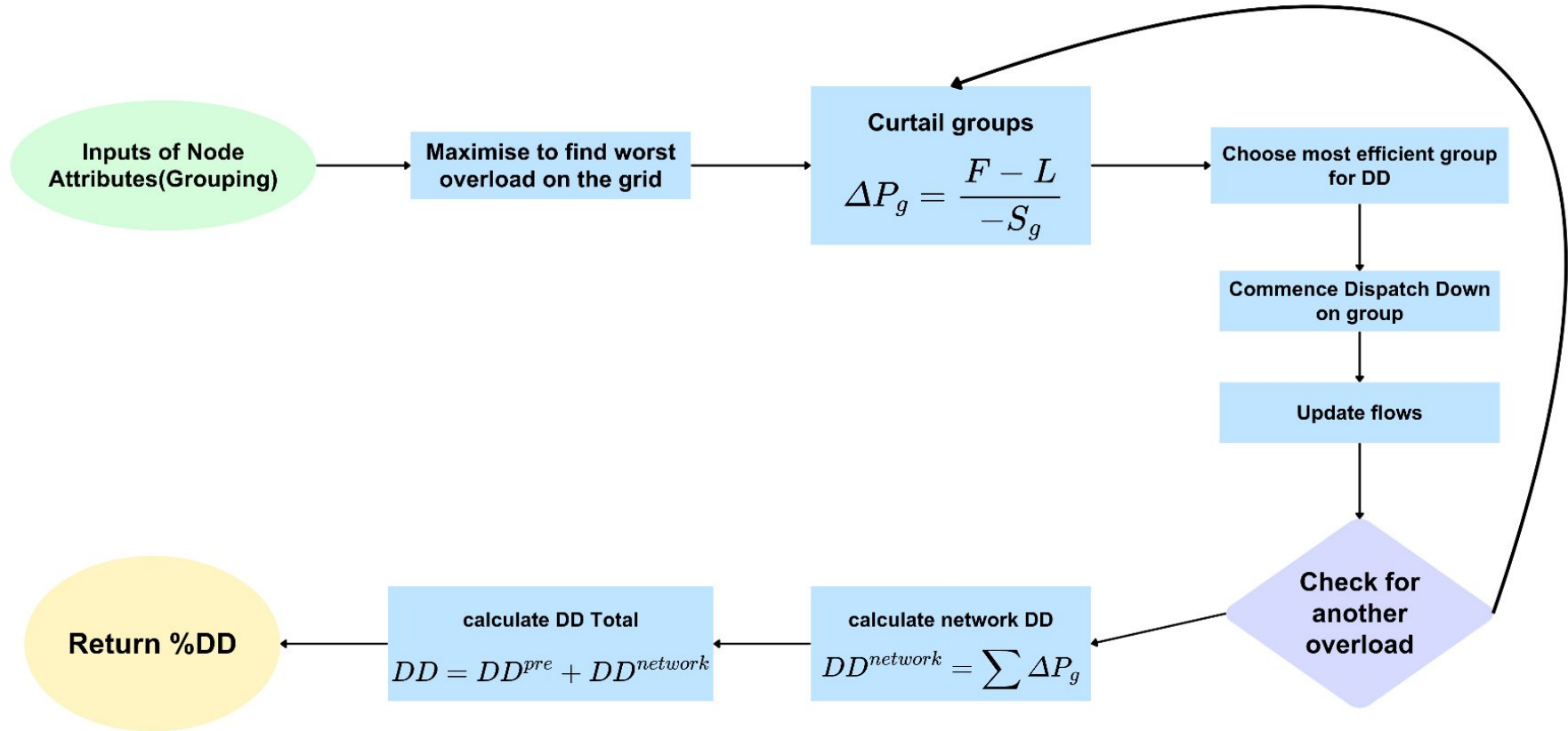
Demand is needed to simulate a surplus in energy as well as a high amount of generation.

We also need some details about the transmission, such as phase angle, voltage, and thermal rating

The initialization of the Simulation, A simplification



The running of the Simulation, A simplification



Assumptions

Dispatch-down reduction is treated as improved renewable utilisation, not electrical-efficiency improvement

No inertia constraints

N-1 only

DC power flow

Interconnectors Ignored*

Shared/common wind profile assumption

Simplified solar daylight profile

Each renewable asset belongs to exactly one constraint group

*the connections between northern ireland and the republic are included

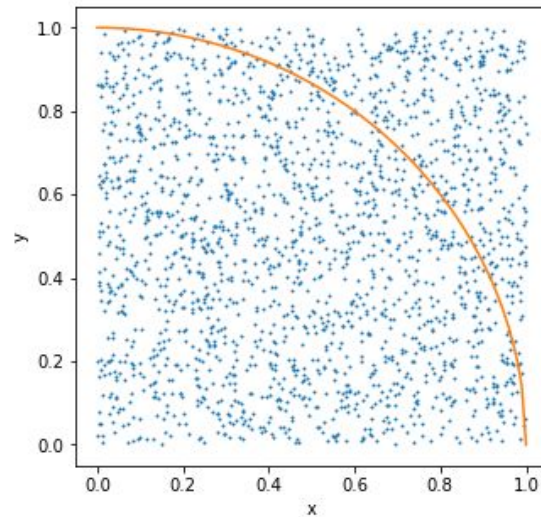
Screened N-1 contingency analysis	DC power flow	10,000 Monte Carlo runs reweight those 168 snapshots
Security screen frozen from the baseline case	Fixed voltage magnitudes	Simplified conventional merit-order dispatch
Security guard is baseline-relative	Reactive power ignored	Fixed conventional minimum output
N-1 only	Small-angle / linear power-flow approximation	No start-up constraints
No dynamic stability modelling	Resistance ignored in the actual power-flow calculation	No minimum up/down-time constraints
No inertia constraints	Thermal MVA ratings approximated as MW limits	Largest conventional generator bus used as the balancing bus
No reserve constraints	Transformer tap-ratio effects omitted	Proportional curtailment within each selected group
No system-services constraints	Transformer phase-shift effects omitted	Each renewable asset belongs to exactly one constraint group
No voltage-security modelling	Only the largest connected AC component is retained	Constraint-group sizes are fixed during annealing
No reactive-power security modelling	HVDC flows are fixed	Total number of constraint groups is fixed
Dispatch-down percentage is the optimisation objective	Synthetic renewable weather profiles	Greedy worst-constraint relief
No ramp-rate constraints	Shared/common wind profile assumption	Maximum number of group-curtailment actions per snapshot
Dispatch-down reduction is treated as improved renewable utilisation, not electrical-efficiency improvement	Simplified solar daylight profile	168 underlying demand snapshots

Simulated Annealing.

Monte Carlo Methods

$$\sigma_n(f(x)) = \frac{1}{n} \sum_i^n f(x^{(i)})$$

$$x^{(i)} \in p(x)$$



Markov Chains

$$P(X_{n+1} = x \mid X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = P(X_{n+1} = x \mid X_n = x_n)$$

“The next state only depend on the previous state, not the history on how we got there”

$$\pi(\sigma)P(\sigma'|\sigma) = \pi(\sigma')P(\sigma|\sigma')$$

Splitting the transition probability $P(\sigma|\sigma')$ into a proposal distribution $g(\sigma'|\sigma)$, the probability of *proposing* σ' given σ , and an acceptance rate $A(\sigma'|\sigma)$ the probability of *accepting* σ' , gives us

$$\pi(\sigma)g(\sigma'|\sigma)A(\sigma'|\sigma) = \pi(\sigma')g(\sigma|\sigma')A(\sigma|\sigma')$$

Detailed balance holds if we define,

$$A(\sigma'|\sigma) = \min \left(1, \frac{\pi(\sigma')g(\sigma|\sigma')}{\pi(\sigma)g(\sigma'|\sigma)} \right)$$

Now inserting the Boltzmann distribution for π

$$\begin{aligned} A(\sigma'|\sigma) &= \min \left(1, e^{-\beta(E(\sigma')-E(\sigma))} \frac{g(\sigma|\sigma')}{g(\sigma'|\sigma)} \right) \\ &= \min \left(1, e^{-\beta\Delta E} \underbrace{\frac{g(\sigma|\sigma')}{g(\sigma'|\sigma)}}_{=1} \right) \end{aligned}$$

$$A(\sigma'|\sigma) = \min \left(1, e^{-\beta\Delta E} \right)$$

1. Choose an initial grouping

$$G^{(0)} = \{g_1^{(0)}, g_2^{(0)}, \dots, g_n^{(0)}\}, \quad g_i^{(0)} \in \{1, \dots, K\}.$$

2. Propose G' by swapping the groups of two nodes,

$$g_i = A, \quad g_j = B \quad \longrightarrow \quad g'_i = B, \quad g'_j = A.$$

3. Define the energy as the simulated dispatch-down,

$$E(G) = D(G).$$

4. Accept G' according to

$$P_{\text{acc}} = \begin{cases} 1, & \Delta E \leq 0, \\ \exp\left(-\frac{\Delta E}{T}\right), & \Delta E > 0, \end{cases} \quad \Delta E = E(G') - E(G).$$

5. Update

$$G^{(m+1)} = \begin{cases} G', & \text{accepted,} \\ G^{(m)}, & \text{rejected.} \end{cases}$$

6. Repeat while cooling,

$$T_{r+1} = \alpha T_r.$$

7. Record the lowest energy found,

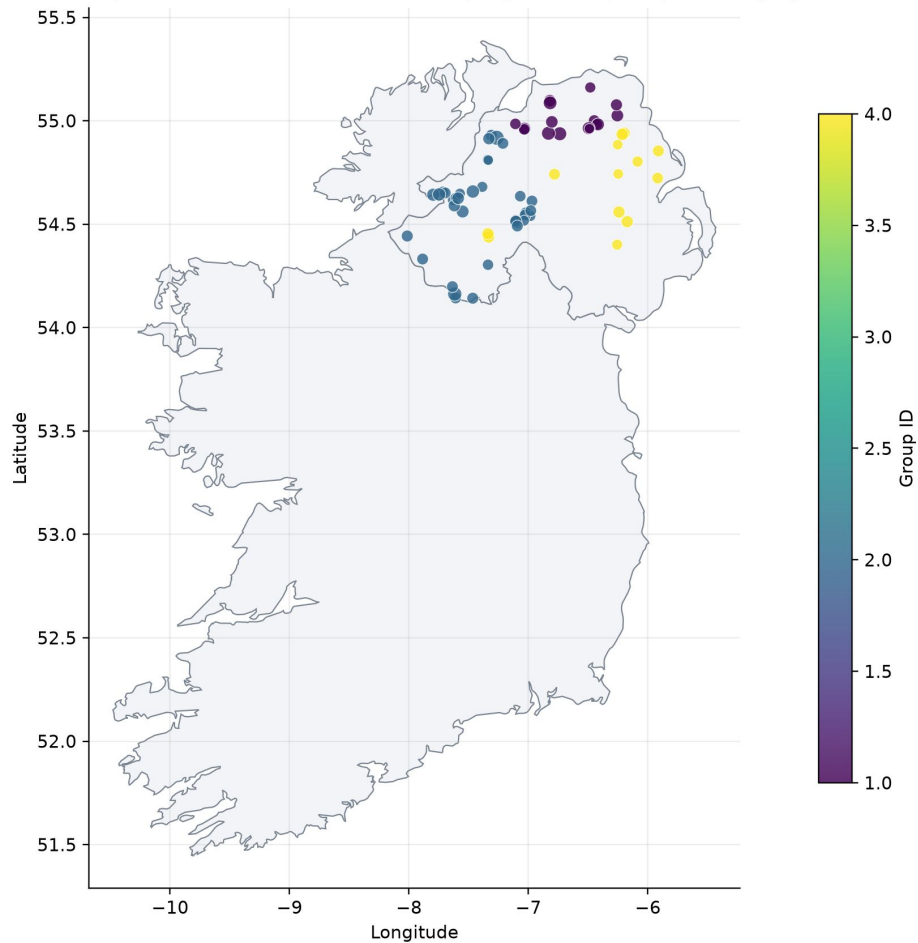
$$E_{\text{best}} = \min_m E(G^{(m)}),$$

and compare it with the baseline grouping.

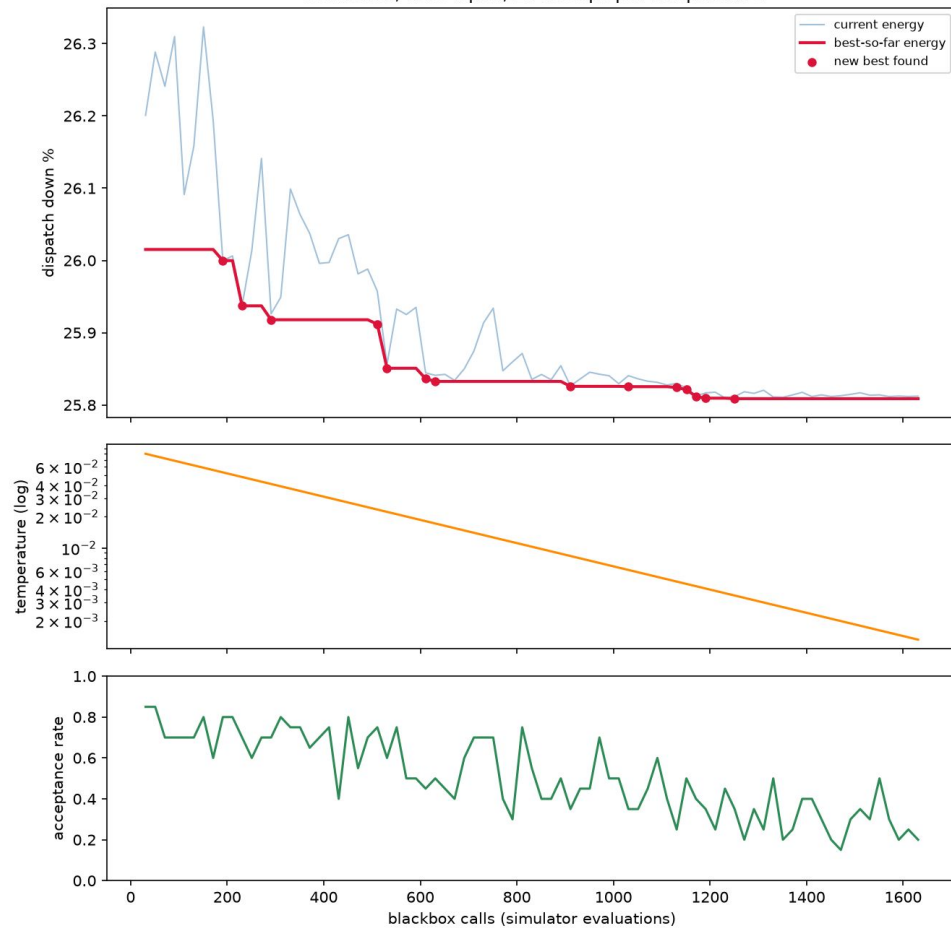
6 counties.

6-county renewable nodes

66 plotted nodes • marker size = renewable capacity • colour = primary/exclusive group



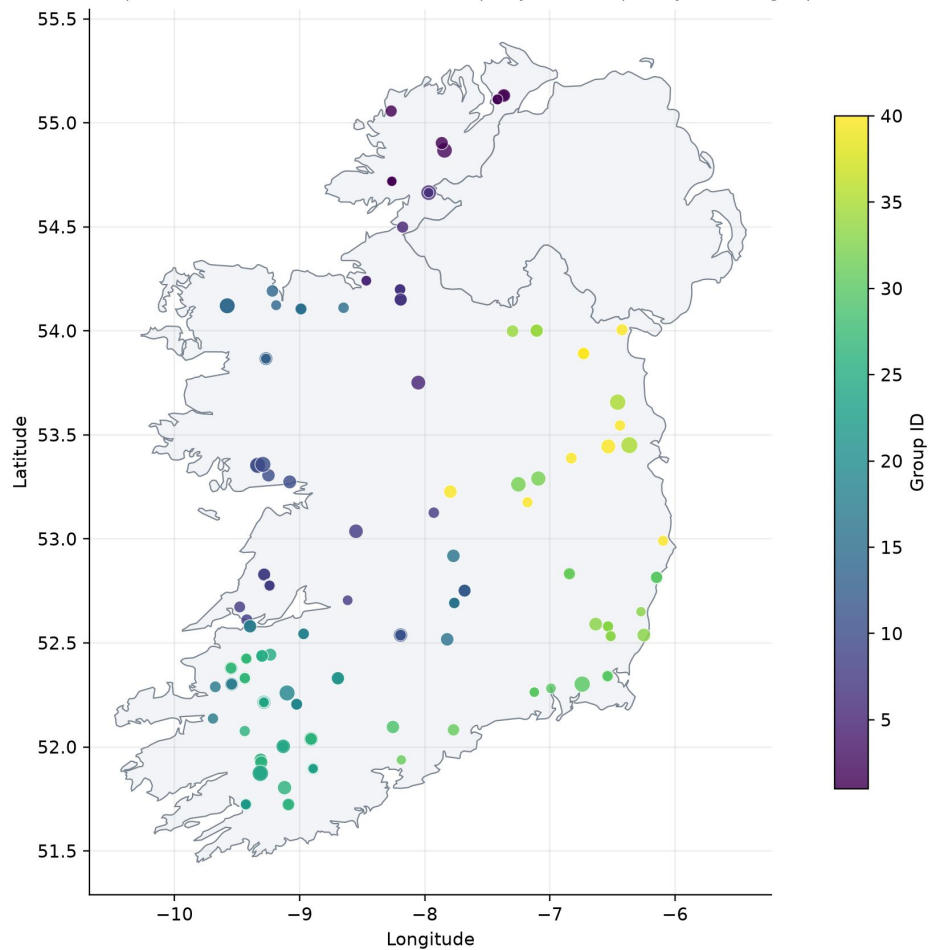
6 counties, 0.95 alpha, 20 sweeps per temperature



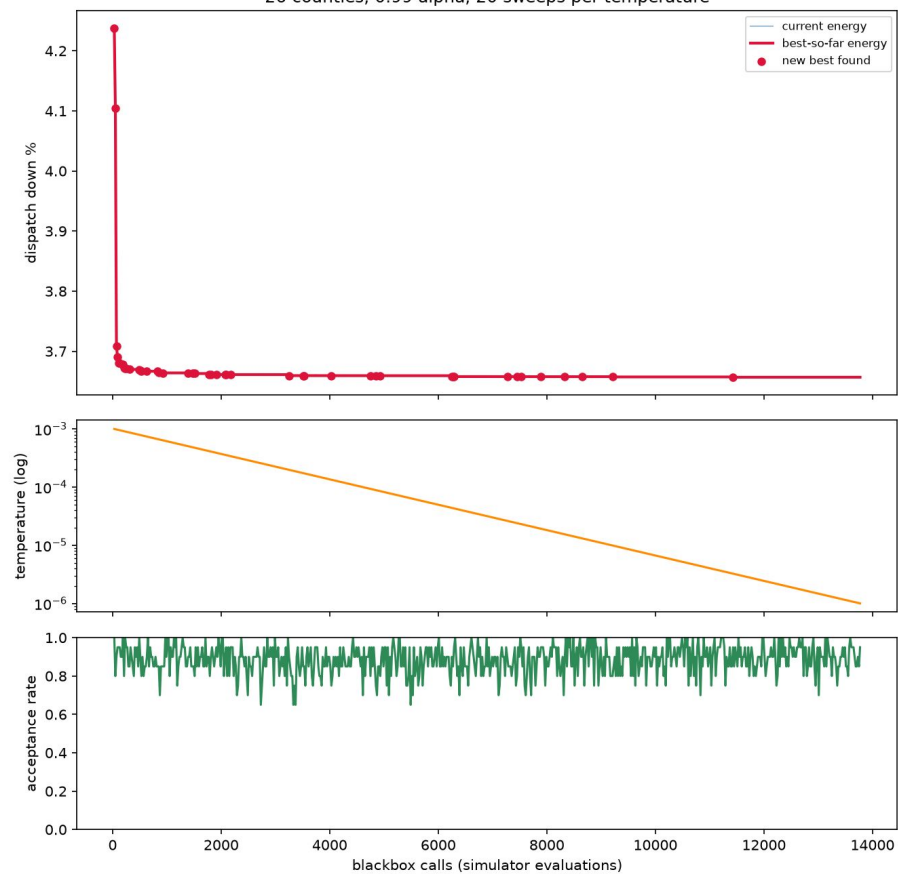
26 counties.

26-county renewable nodes

174 plotted nodes • marker size = renewable capacity • colour = primary/exclusive group



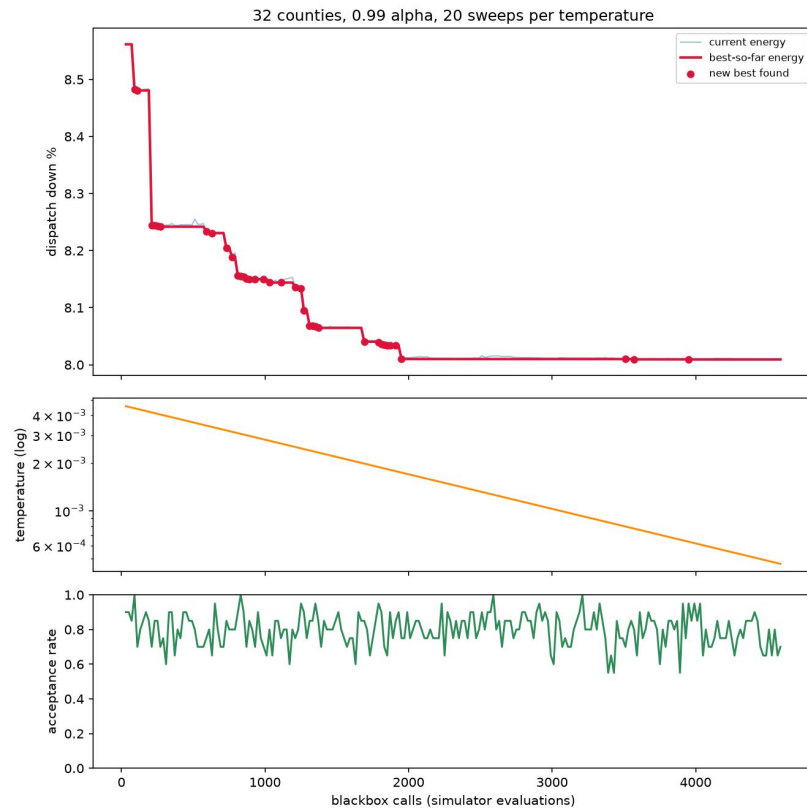
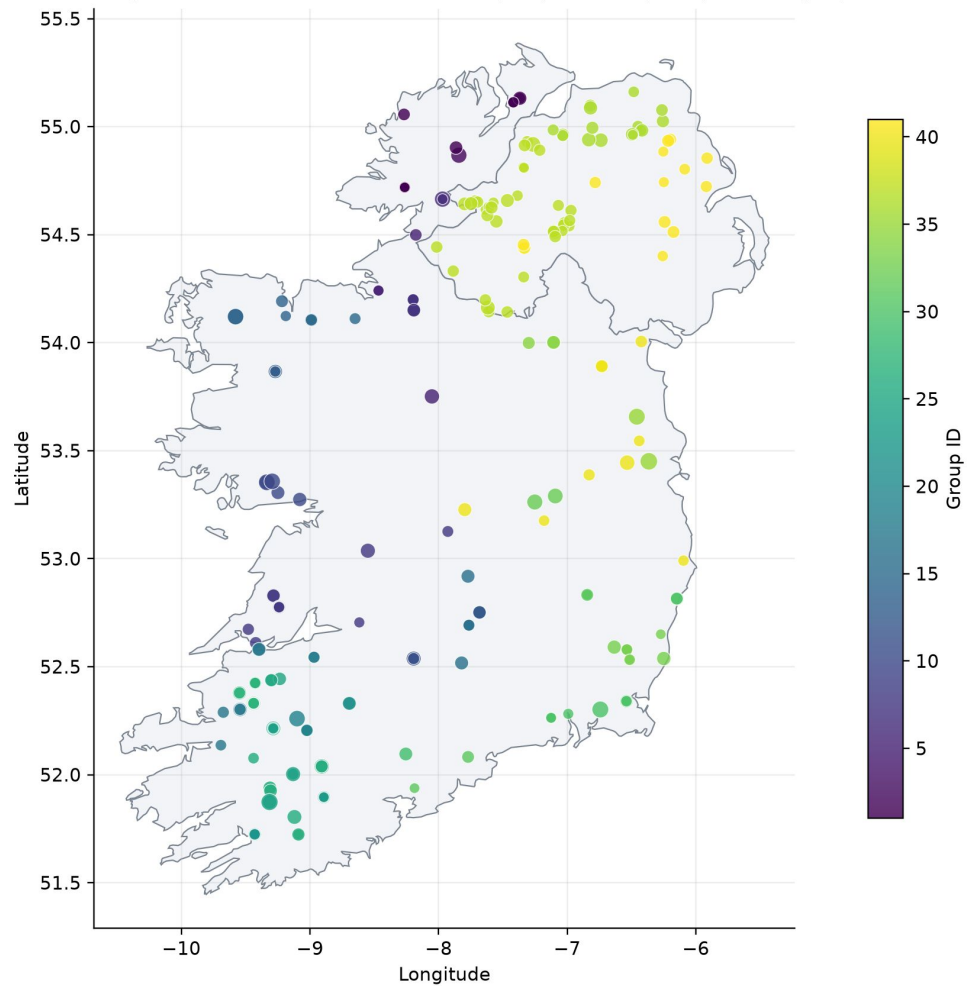
26 counties, 0.99 alpha, 20 sweeps per temperature



And we unite
Ireland.

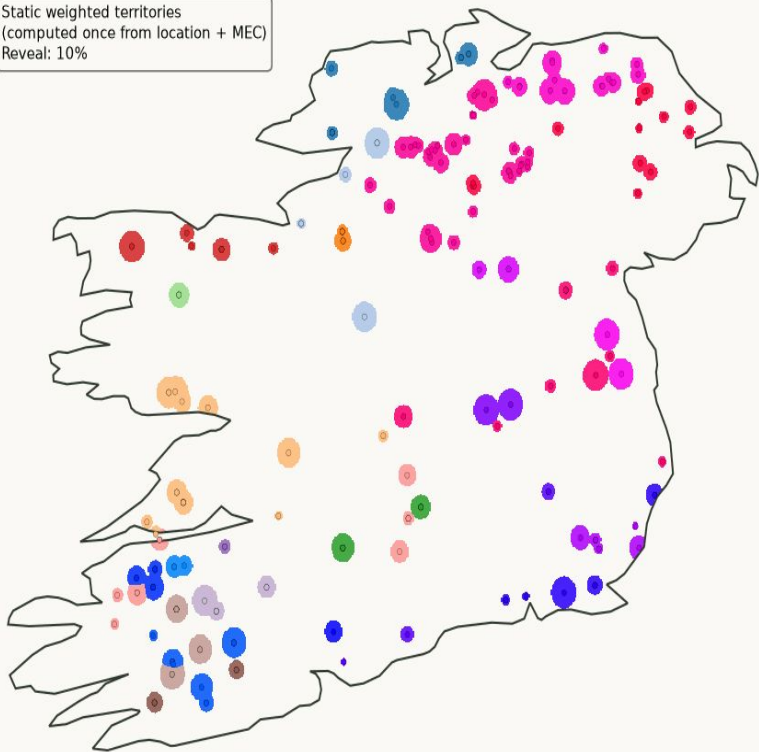
32-county renewable nodes

240 plotted nodes • marker size = renewable capacity • colour = primary/exclusive group



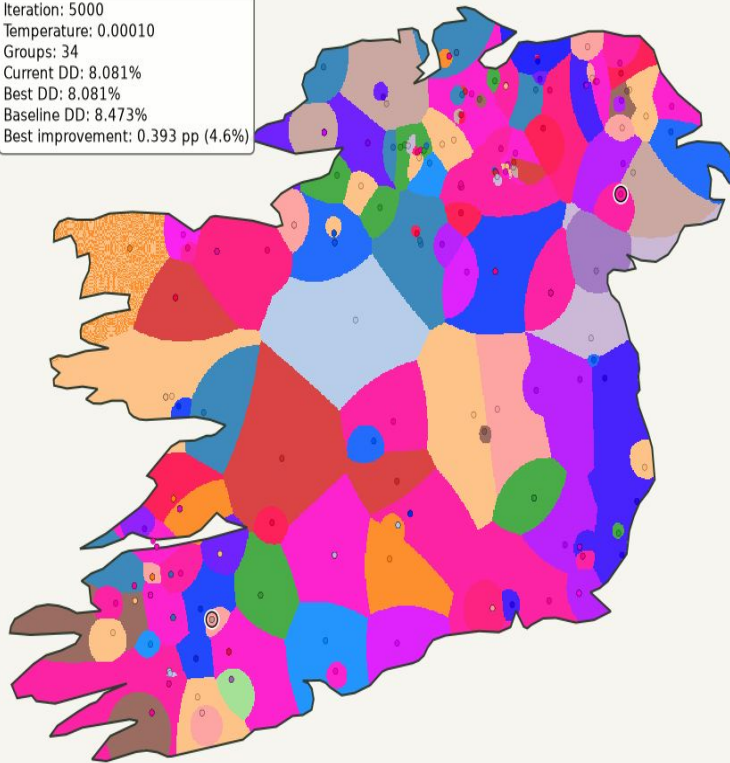
32-county annealing territories

Static weighted territories
(computed once from location + MEC)
Reveal: 10%



32-county annealing territories (final)

Iteration: 5000
Temperature: 0.00010
Groups: 34
Current DD: 8.081%
Best DD: 8.081%
Baseline DD: 8.473%
Best improvement: 0.393 pp (4.6%)

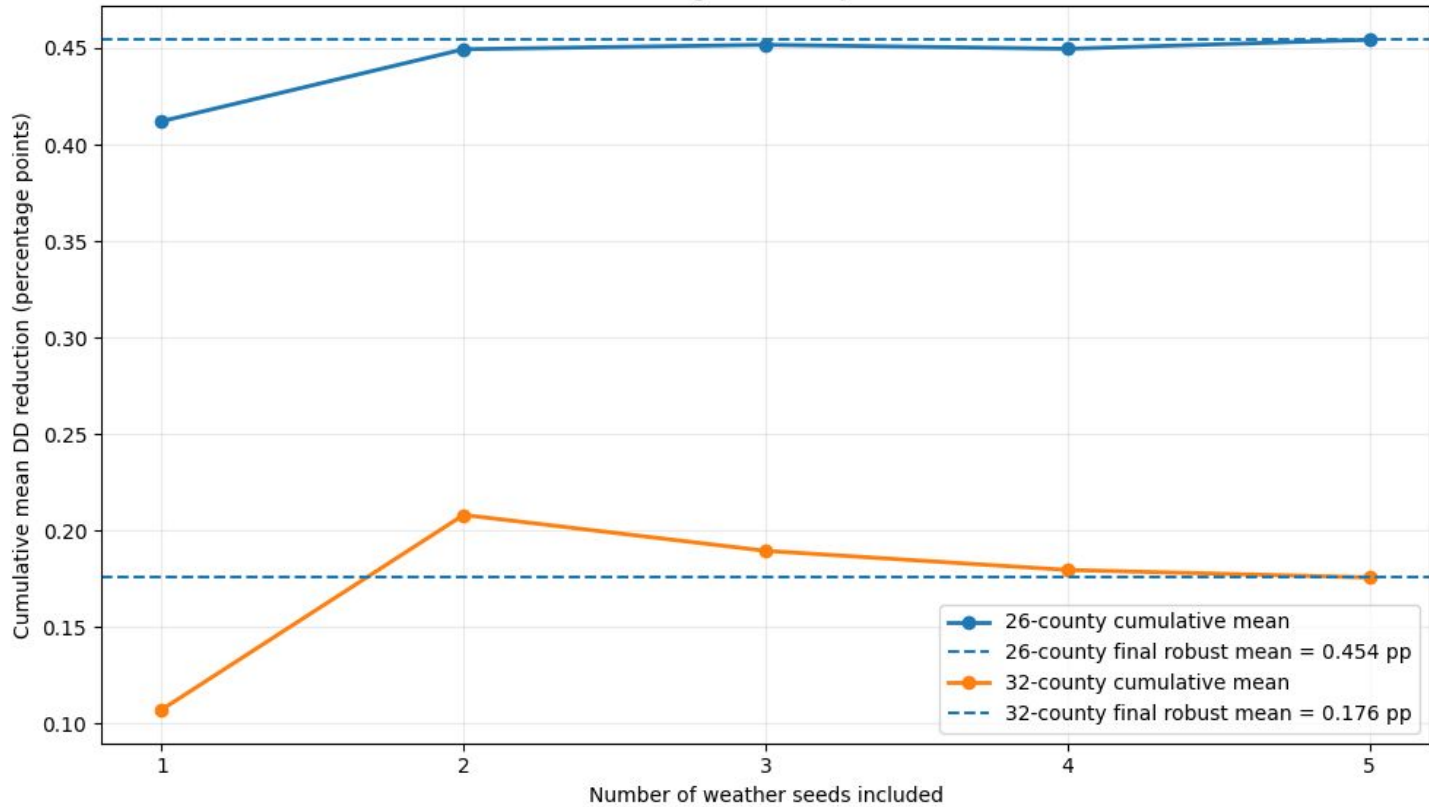


Verification.

- Multiple weather seeds are tested
- Multiple simulation seeds used
- Baseline and optimized groups are always compared under the same conditions
- N-1 contingencies are included.
- Stress scenarios are added.
- Simulator ran 5000 times (and then each sim has 1000 iterations-ish) before any decisions on the rigidity of the system are made.

- **26-county candidate:** 0.454 percentage point mean reduction, equivalent to 7.3% relative improvement, safe across all 5 nominal N-1 weather-seed checks.
- **32-county candidate:** 0.176 percentage point mean reduction, equivalent to 1.9% relative improvement, safe across all 5 nominal N-1 weather-seed checks.

Cross-weather convergence of dispatch-down reduction



Thank You

github.com/ciangregg/EIEG_Hackathon26

Questions?